1 **TRUE/FALSE** INCORRECT

The Turing Test is a method of inquiry in artificial intelligence (AI) for determining whether a computer is capable of thinking like a human being

- True *Correct answer*
- False

2 **TRUE/FALSE** INCORRECT

Ethical issues related to AI include privacy concerns and the impact on job markets.

- True *Correct answer*
- False

3 **TRUE/FALSE** INCORRECT

If we interpret the predicate **Spouse(x,z)** as "x is the Spouse of z", the following FOL formula means "Everyone can be the spouse of any given person."

$\exists x \forall$ x,y,z Spouse(x,z) $\wedge$ Spouse(y,z) $\Rightarrow$ x=y

- True
- False *Correct answer*

4 **TRUE/FALSE** INCORRECT

Ethical issues related to AI include privacy concerns and the impact on job markets.

- True *Correct answer*
- False

5 **TRUE/FALSE** INCORRECT

Move ordering has no effect on Alpha-Beta pruning

- True
- False *Correct answer*

6 **TRUE/FALSE** INCORRECT

The minimax algorithm is only applicable to two-player, zero-sum games.

- True *Correct answer*
- False

Which among the following is not a primary characteristic of an intelligent agent?

A. Reactivity

B. Ability to play board games — Correct answer

C. **Incorrect:** Proactiveness

D. Autonomy

8 **MULTIPLE CHOICE** INCORRECT

What is the CNF form for the following sentence?
(P => P) <=> Q

A. (P v ¬P v ¬Q) ^ (¬P v Q) ^ (P v Q) — Correct answer

B. (¬P v ¬Q) ^ (¬P v Q) ^ (P v Q)

C. (P ^ ¬Q) v (¬P ^ Q) v (P ^ Q)

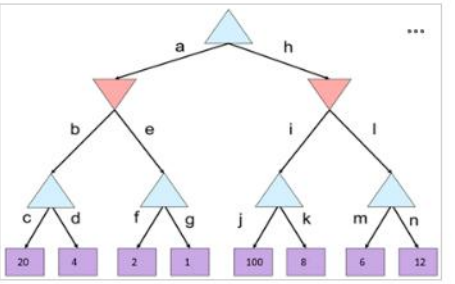
D. (P ^ ¬P) v (¬Q ^ ¬P) v (Q ^ P)

9 **MULTIPLE CHOICE** INCORRECT

The CNF (Clausal) form of A->(B ^C) is

- A. (AVB)^ (~AVC)
- B. (~AV~B)^ (AVC)
- C. (AV~B)^(AV~C)
- D. (~AVB)^(~AVC) Correct answer

The following game tree generated using the MiniMax algorithm.
What will be the value in the root node?



A. 100

B. 2

C. **Correct:** 12 — Correct answer

D. 20

11 **MULTIPLE CHOICE** INCORRECT

Which of the following is an Informed Search technique?

- A. Breadth first search
- B. Depth first search
- C. Depth limited search
- D. **Correct:** A* algorithm — Correct answer

12 **MULTIPLE CHOICE** INCORRECT

A knowledge base has the following rules

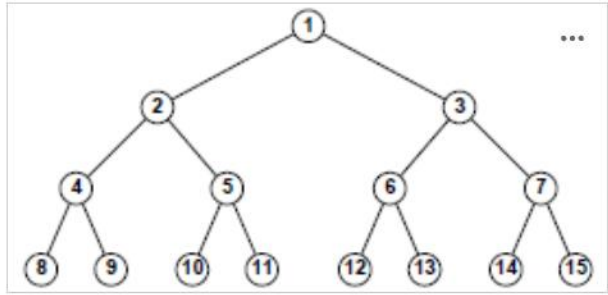
1. P->Q 2. L^ M->P, 3. B^L->M , 4. A^B->L, 5. A, 6. B

If the query Q has to be tested using backward chaining inference method, what sequence of the rule will be executed.

- A. 5,6,4,2,3,1
- B. **Correct:** 1,2,4,5,6,3 — Correct answer

13 **MULTIPLE CHOICE** INCORRECT

What is the sequence of nodes visited during Depth First Search?



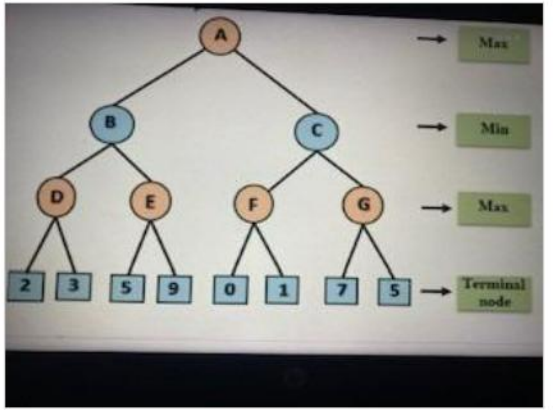
A. 1,2,3,4,5,6,7,8,9,10,11,12,13,14,15

B. **Correct:** 1,2,4,8,9,5,10,11,3,6,12,13,7,14,15 Correct answer

C. 1,2,4,9,8,5,11,10,3,6,13,12,7,15,14

D. 13,6,12,13,7,14,15,2,4,8,9,5,10,11

IF alpha-beta pruning is used in the following graph, which of the following gives the alpha and beta values of node (E)



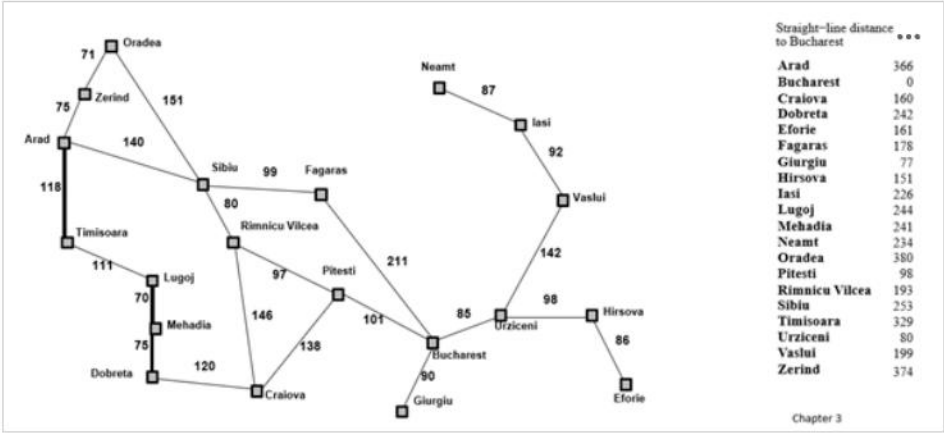
A. 2 , 5

B. 3 , 5

C. **Correct:** 5 , 3 — Correct answer

D. 3 , 9

A* is applied to the problem of getting from Craiova to Bucharest using the straight-line distance heuristic. What is the cost for Pitesti after the application of the evaluation function f(x) = g(x) + h(x) ?



Your answer: 236

1 **TRUE/FALSE** INCORRECT

Learning in Neural Networks happens through the adjustment of weights

- True *Correct answer*
- False

2 **TRUE/FALSE** INCORRECT

In computer vision implementations, classification is more difficult than segmentation

- True
- False *Correct answer*

3 **TRUE/FALSE** INCORRECT

ChatGPT uses LSTM models

- True
- False *Correct answer*

4 **TRUE/FALSE** INCORRECT

Generative AI models learn the patterns and structure of their input training data and then generate new data that has similar characteristics.

- True *Correct answer*
- False

Identify the type of learning in which only labelled training data is used.

A. Unsupervised learning

B. Reinforcement learning

C. Semi supervised learning

D. **Correct:** Supervised learning — Correct answer

6 **MULTIPLE CHOICE** INCORRECT

You are asked to implement a neural network that classifies samples into 5 groups. Each sample is represented by 3 values. How would you design your neural network?

A. **Correct:** Input layer with 3 inputs, output layer with 5 outputs and multiple hidden layers. — Correct answer

B. Input layer with 5 inputs, output layer with 3 outputs and multiple hidden layers.

C. Input layer with 2 inputs, output layer with2 outputs, and multiple hidden layers,

Select the primary purpose of large language models (LLMs)

A. To create a new language

B. **Correct:**
To process and understand vast amounts of natural language data

Correct answer

C. To translate languages in real-time

D. To summarize text

8 **MULTIPLE CHOICE** INCORRECT

You have a balanced dataset with 1000 samples. Which of the following data split strategies are correct?

A. 50% train, 50% test

B. 95% train, 5% test

C. **Correct:**
80% train, 10% validation, 10% test

Correct answer

D. 20% train, 10% validation, 70% test

Last attempt with a mark

9 MULTIPLE CHOICE INCORRECT

In a CNN model the Input volume is 32x32x3. Applying 10 5x5 filters with stride 1 , padding 2 what is the output feature volume?

A. 36x36x10

B. 34x34x10

C. **Correct:** 32x32x10 — Correct answer

D. 30x30x10

10 MULTIPLE CHOICE INCORRECT

In question 9, how many parameters in this convolutional layer?

A. **Correct:** 760 — Correct answer

B. 750

C. 1024

Which of the following is the primary purpose of convolutional layers in a Convolutional Neural Network (CNN)?

A. To reduce the dimensionality of the data using pooling techniques

B. To extract spatial features such as edges, textures, and patterns — Correct answer

C. To connect every neuron to every other neuron in adjacent layer

D. **Incorrect:**
To classify input data into predefined categories

2 **MULTIPLE CHOICE** INCORRECT

What is the primary function of RELU activation function in a neural network?

A. **Incorrect:**
To update the weights during backpropagation

B. To initialize the weights of the network

C. To calculate the loss between predicted and actual outputs

D. To introduce non-linearity into the model and help it learn complex patterns — Correct answer

What is a key characteristic of a Large Language Model (LLM)?

A. It operates exclusively on structured data such as tables and graphs.

B. It requires manual feature engineering for every natural language task.

C. **Correct:**
It uses billions of parameters to understand and generate human-like text. Correct answer

D. It is trained only on domain-specific datasets for niche applications.

14 **MULTIPLE CHOICE** INCORRECT

Which of the following techniques is commonly used to fine-tune a Large Language Model (LLM) for a specific task?

A. **Incorrect:**
Convolutional layers for feature extraction

B. Hard-coding rules for natural language processing

C. Using decision trees to interpret the model outputs

D. Transfer learning with task-specific data Correct answer

15 **MULTIPLE CHOICE** INCORRECT

Which of the following loss functions is most commonly used for multi-class classification problems?

- A. Categorical Cross-Entropy loss — Correct answer
- B. **Incorrect:** Mean Squared Error (MSE)
- C. Mean Absolute Error (MAE)
- D. Cosine similarity